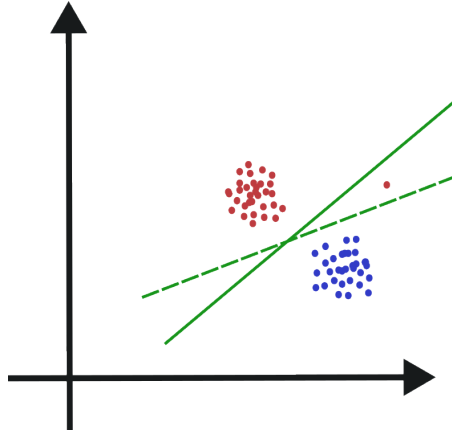


Machine Learning Exercise 2

EXERCISE 1

Construct (on paper) an example for a 2-dimensional dataset which is linearly separable, but for which linear discriminant analysis (LDA) will not define decision regions that separate the data. To answer this question it is only important to understand the underlying principles of LDA, not the exact computations by which an LDA model is learnt.

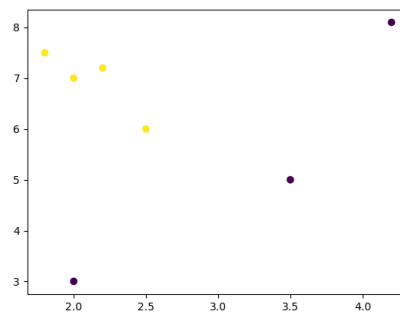
Solution:



The red and blue datapoints are separable by the dashed line. However, LDA will learn a decision boundary close to the solid green line: the impact of the red outlier point on the learned parameters (slide 2.9) is negligible, and the learned model will consist of two near perfectly “round” Gaussians (i.e., with zero covariance) centered on the means of the red and blue clusters. This leads to the solid decision boundary.

EXERCISE 2 Here is a very small dataset with two classes:

X_1	X_2	Y
2.	3.	0
3.5	5.	0
4.2	8.1	0
2.	7.	1
2.2	7.2	1
2.5	6.	1
1.8	7.5	1



Compute the mean vectors μ_0, μ_1 and covariance matrices Σ_0, Σ_1 for the two classes $Y = 0, 1$ (cf. slide 2.9).

Solution:

Using the formulas from slide 2.9:

$$\begin{aligned}\mu_0 &= (3.233, 5.366) \\ \mu_1 &= (2.125, 6.925) \\ \Sigma_0 &= \begin{pmatrix} 1.26 & 2.73 \\ 2.73 & 6.6 \end{pmatrix} \\ \Sigma_1 &= \begin{pmatrix} 0.09 & -0.174 \\ -0.174 & 0.422 \end{pmatrix}\end{aligned}$$

This can be done with pencil and paper, but a little computational support from a calculator, spreadsheet, Python, ... can be helpful.

EXERCISE 3

Consider the following dataset. The data has one numeric attribute X_1 , a categorical attribute X_2 with three possible states $\{a, b, c\}$, and a binary class label Y .

X_1	X_2	Y
25	a	0
2	a	1
1	b	0
-5	b	0
2	c	1
2	b	0

Find parameters for a logistic regression model that classifies these 6 datapoints correctly (there are many solutions).

Solution:

The logistic regression model uses a one-hot-encoding for the categorical variable X_2 , so the data effectively looks like this:

X_1	$X_{2,a}$	$X_{2,b}$	$X_{2,c}$	Y
25	1	0	0	0
2	1	0	0	1
1	0	1	0	0
-5	0	1	0	0
2	0	0	1	1
2	0	1	0	0

We then need to find parameters $w_0, w_1, w_{2,a}, w_{2,b}, w_{2,c}$ so that $\mathbf{w}\mathbf{x} \geq 0$ for the cases with $Y = 1$. We first observe that whenever $X_{2,b} = 1$ we have $Y = 0$. So we can safely let $w_{2,b}$ be a large negative weight, say $w_{2,b} = -100$. Similarly, we can set $w_{2,c} = +100$. Finally, we need to separate the first two datapoints. This we can do by setting $w_1 = -1$ and $w_0 = 5$. The added term $w_0 + w_1 X_1$ does not change the sign of the function value for the last four points, since it is much less than 100 in absolute value.

EXERCISE 4

Consider again our XOR example, where we want to classify 4 datapoints as follows:

A	B	Class
1	1	$\ominus$
1	0	$\oplus$
0	1	$\oplus$
0	0	$\ominus$

Define a feature function $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^D$ (for some suitable choice of D), so that the 4 datapoints become linearly separable in the feature space.

Solution:

For example we can use $D = 1$ and define $\phi(A, B) = (A - B)^2$.